

YOUSI FU

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Education

Bioengineering,	B.E.	Dalian University of Technology, China
Bioengineering,	M.E.	Dalian University of Technology, China
Biochemical Engineering,	Ph.D.	Xiamen University, China

Work Experience

Sep 2019 to Nov 2021	Postdoc	Xiamen University, China
June 2022 to June 2026	Research Associate	Michigan State University, USA
June 2026 to Now	Associate Professor	Nankai University, China

Research Background

Microbial Bile Acids Metabolism

- Elucidated the mechanism by which conjugated bile acids, especially microbial conjugated bile acids (MCBAs), modulate microbiome, metabolism, and immunity, demonstrating their therapeutic efficacy in a murine ulcerative colitis model.
- Characterized of bile salt hydrolase (BSH) in inflammatory bowel disease (IBD) via machine learning analysis of metagenomic sequencing data.
- Investigate how *Ruminococcus gnavus* BSH catalyzes MCBAs synthesis and its role in IBD development.

Microbiome-Targeted Intervention Strategies

- Designed a novel prebiotic formulation (*Dendrobium officinale* polysaccharide and berberine) that selectively enriches short chain fatty acids-producing microbiota and optimizes bacterial communities *in vitro*, validated through multi-omics integrations.
- Established the *in vitro* cultured human microbiota for fecal transplantation therapy (FMT) in mice modeling autism spectrum disorder (ASD), identifying gut-brain axis-mediated neuroprotective effects of specific microbiome on ASD symptoms.
- Created a multi-task deep learning framework that deciphers cross-cancer associations between gut microbiome features and PD-1 immunotherapy response.
- Engineering a high-throughput and affordable platform for gut microbiota *in vitro* culture and applying it to microbial culturomics studies.

Bioactive Peptide Discovery & Translational Applications

- Pioneered LC-MS/MS-guided bioactivity screening to isolate novel antioxidant collagen peptides from bovine bone enzyme hydrolysates.
- Developed machine learning models to predict antioxidant peptides, enabling efficient screening of bovine bone enzyme hydrolysates.
- Identification of two lifespan-extending collagen peptides in *Caenorhabditis elegans*.
- Screening and characterization of two collagen peptides for osteoporosis intervention in the MC3T3-E1 cell model.

Anaerobic Bioprocess Engineering

- Acetone-butanol-ethanol (ABE) fermentation by *Clostridium acetobutylicum*.
- Investigated the industrial production and product separation of 1,3-propanediol via *Clostridium butyricum* fermentation.

Teaching

Mentorship

- Supervised 2 undergraduate and 3 master's students in microbiome-related and peptide biochemistry projects, providing training in molecular biology, enzyme assays, and data analysis.
- Served as advisor for the XMU-China iGEM team, leading interdisciplinary groups of undergraduates in synthetic biology project design, experimental planning, and presentation development.

Teaching assistant

- Bioprocess Modeling, Xiamen University (Ph.D. program)
- Assisted in course delivery for graduate-level training in quantitative modeling of biological systems.

Awards

- 2025 *Journal of Future Foods*-Outstanding Reviewer Certificate.
- 2024 MDPI Top 1000 Reviewers.
- 2025 Outstanding Postdoctoral Scholar Award in Department of Biochemistry and Molecular Biology, MSU.
- From 2017 to 2020, won the gold medal of International Genetically Engineered Machine Competition (iGEM) for four consecutive years (as the deputy PI or advisor of the XMU-China team).
- 2019 Outstanding PhD Graduate Awards of Xiamen University.
- 2016 and 2019 Excellent Student Awards of Xiamen University.
- 2013 Outstanding Graduate Awards and Excellent Undergraduate Thesis Awards of Dalian University of Technology.
- 2010 and 2011 Dalian University of Technology Study Scholarship.

Presentations

2026 Genetic conservation, dual functionality, and potential role of *Ruminococcus gnavus* bile salt hydrolase in inflammatory bowel disease, Offered talk, Microbes in Microbiomes, Miami, FL.

2024 The roles of bile acid conjugation in gut inflammation, Oral Presentation, Anaerobe at the University of Michigan, MI.

Service

Reviewer for *Journal of Nanobiotechnology*, *Communications Biology*, *Journal of functional foods*, *Food Bioscience*, *Metabolites*, *International Journal of Molecular Sciences*, and *Nutrients* journals.

Publications

Microbial Bile Acids Metabolism

- [1] **Yousi Fu**, Douglas V. Guzior, Maxwell Okros, Christopher Bridges, Sabrina L. Rosset, Cely T. González, Christian Martin, Hansani Karunarathne, Victoria E. Watson, and Robert A. Quinn*; Balance between bile acid conjugation and hydrolysis activity can alter outcomes of gut inflammation. *Nature Communications*, 2025, 16, 3434.
- [2] Tae Hyung Won, Mohammad Arifuzzaman, Christopher N Parkhurst, Isabella C Miranda, Bingsen Zhang, Elin Hu, Sanchita Kashyap, Jeffrey Letourneau, Wen-Bing Jin, **Yousi Fu**, Douglas V Guzior, Robert A Quinn, Chun-Jun Guo, Lawrence A David, David Artis*, Frank C Schroeder*; Host metabolism balances microbial regulation of bile acid signalling. *Nature*, 2025, 638, 216-224.
- [3] Douglas V. Guzior, Maxwell Okros, Madison Shivel, Bruin Armwald, Christopher Bridges, **Yousi Fu**, Christian Martin H., Anthony Schillmiller, Wendy M. Miller, Kathryn M. Ziegler, Matthew D. Sims, Michael E. Maddens, Stewart F. Graham, Robert P. Hausinger, and Robert A. Quinn*; Bile salt hydrolase acyltransferase activity shapes bile acid diversity and biochemistry. *Nature*, 2024, 626:852-858.

Microbiome-Targeted Intervention Strategies

- [4] **Yousi Fu**, Yali Wang, Junnan Zhang, Jianlin Ren* and Baishan Fang*, Prebiotic structural diversity shapes gut microbial diversity, community composition, and metabolic activity *in vitro*, *Foods*, 2025, 14, 3774.
- [5] Cely T González, Christian Martin, Maddey Crane, Karen Gutierrez, Jacob Thomas, Lacy Remisoski, Maxwell Okros, **Yousi Fu**, Douglas V Guzior, Dustin Finkhouse, Christopher Bridges, Jenna Mielke, Gabriel Querido, Lienwil Padillo, Reda Girgis, Marc McClelland, Douglas Conrad, Xiaopeng Li, Robert A Quinn*, Microenvironmental Effects of a Non-Antibiotic Therapy for a Chronic Polymicrobial Infection Alter Microbial Physiology, Competition, and Virulence, *The ISME Journal*, 2025, wraf125.
- [6] **Fu Yousi**[#], Wang Yali[#], Sun Yidan, Ren Jianlin* and Fang Baishan*; Responses of human gut microbiota abundance and amino acid metabolism in vitro to berberine; *Food & Function*, 2022, 13, 6329-6337. (#equal contribution)
- [7] Jiang Changzhi, **Fu Yousi**, Jin Shuting, Liu Xiangrong, Fang Baishan, Zeng Xiangxiang*. A multi-task learning method for analyzing microbiota as cancer immunotherapy signal. IEEE International Conference on Bioinformatics and Biomedicine (**BIBM**); IEEE, 2020, 437-441.
- [8] Chen Kainan[#], **Fu Yousi**[#], Wang Yali, Liao Langxing, Xu Hongzhi, Zhang Aihui, Zhang Junnan, Fan Lina, Ren Jianlin*, Fang Baishan*; Therapeutic Effects of the In Vitro Cultured Human Gut Microbiota as Transplants on Altering Gut Microbiota and Improving Symptoms Associated with Autism Spectrum Disorder; *MICROBIAL ECOLOGY*, 2020, 80(2):475-864. (#equal contribution)
- [9] Luxi Yang, Kaijian Hou, Bangzhou Zhang, Cong Ouyang, Aiqiang Lin, Shuangbin Xu, Dongxian Ke, Lujing Fang, Qiongyun Chen, Jingtong Wu, Changsheng Yan, Yifan Lian, Tao Jiang, Jianquan He, Han Wang, **Yousi Fu**, Chuanxing Xiao, Zhangran Chen*, Preservation of the fecal samples at ambient temperature for microbiota analysis with a cost-effective and reliable stabilizer EffoGut, *Science of The Total Environment*, 2020, 41:140423.
- [10] **Fu Yousi**, Zhang Junnan, Chen Kainan, Xiao Chuanxing, Fan Lina, Zhang Bangzhou, Ren Jianlin*, Fang Baishan*; An in vitro fermentation study on the effects of Dendrobium officinale polysaccharides on human intestinal microbiota from fecal microbiota transplantation donors; *Journal of Functional Foods*, 2019, 53:44-53.
- [11] **Fu Yousi**, Chen Kainan, Zhang Junnan, Xiao Chuanxing, Fan Lina, Zhang Bangzhou, Ren Jianlin*, Fang Baishan*; Evaluation of the effects of four media on human intestinal microbiota culture in vitro; *AMB EXPRESS*, 2019, 9:69.

Bioactive Peptide Discovery & Translational Applications

- [12] Yali Wang, Jiaqing Yang, Baishan Fang, Kangxu Wang*, **Yousi Fu***, An AI-driven multilayer strategy and curated dataset for mining antioxidant peptides from yak bone collagen hydrolysates; *Food Research International*, 2026: 118689.
- [13] Yali Wang, Yue Wang, Baishan Fang*, and **Yousi Fu***. *In Silico* Screening and Identification of Functional Peptides from Yak Bone Collagen Hydrolysates: Antioxidant and Osteoblastic Activities; *International Journal of Molecular Sciences*, 2025, 26, 4570.
- [14] Yali Wang, Yue Wang, Xiaoyan Zhuang, Yonghui Zhang, Baishan Fang*, and **Yousi Fu***. Unraveling the Osteogenic Activity and Molecular Mechanism of an Antioxidant Collagen Peptide in MC3T3-E1 Cells; *Nutrients*, 2025, 17(5):824.
- [15] Wang Yali, Sun Yidan, Wang Xingguo, Wang Yue, Liao Langxing, Zhang Yonghui, Fang Baishan*, and **Fu Yousi***; Novel antioxidant peptides from Yak bones collagen enhanced the capacities of antiaging and antioxidant in *Caenorhabditis elegans*; *Journal of Functional Foods*, 2022, 89: 104933.
- [16] Song Yihang[#], **Fu Yousi**[#], Huang Shiyang, Liao Langxing, Wu Qian, Wang Yali, Ge Fuchun, Fang Baishan*; Identification and antioxidant activity of bovine bone collagen-derived novel peptides prepared by recombinant collagenase from *Bacillus cereus*; *Food chemistry*, 2021, 349:129143. (#equal contribution)

Anaerobic fermentation engineering

- [17] Zhang Aihui, Liu Haolin, Huang Shiyang, **Fu Yousi**, Fang Baishan*; Metabolic profiles analysis of 1,3-propanediol production process by Clostridium butyricum through repeated batch fermentation coupled with activated carbon adsorption; *Biotechnology and Bioengineering*, 2018, 115:684-693.
- [18] **Fu Yousi**, Wu Youduo, Chen Lijie*. Synergic effect of Zn²⁺, Ca²⁺, Mn²⁺ on acetone-butanol-ethanol fermentation; *Chemical*

